



Batterjee Medical College
For Science and Technology

Pleural Diseases

An approach to a Case of Chest Pain

Dr. Essam Kotb
Associate professor
Internal Medicine Department

Learning Objectives

Knowledge:

- Describe the normal anatomy of the visceral and parietal pleura and the pathophysiology of fluid or air accumulation in the pleural space.
- Differentiate between transudative and exudative pleural effusions based on underlying causes.
- Explain the classification, risk factors, and underlying mechanisms of spontaneous, traumatic, iatrogenic, and tension pneumothorax.
- Detail Light's criteria and pleural fluid parameter interpretations used to evaluate effusions.
- Outline evidence-based management strategies for transudative effusions, complicated vs. uncomplicated parapneumonic effusions, empyema, and pneumothorax.
- Differentiate between obstructive, non-obstructive and traumatic/spontaneous pneumothorax mechanisms

Skills:

- Identify and interpret the clinical signs, symptoms, and physical exam findings of pleural diseases.
- Analyze diagnostic tests, including chest X-rays, ultrasound, CT scans, and pleural fluid lab results, to formulate differential diagnoses.
- Apply clinical algorithms to choose appropriate procedures based on patient stability.
- Recognize life-threatening conditions like tension pneumothorax and select immediate intervention steps over diagnostic delay.

Values:

- Demonstrate a professional attitude by adhering to clinical training rules, institutional protocols, and established guidelines.
- Value evidence-based practice and patient safety when deciding between conservative observation and invasive procedures.

Main Subtopics

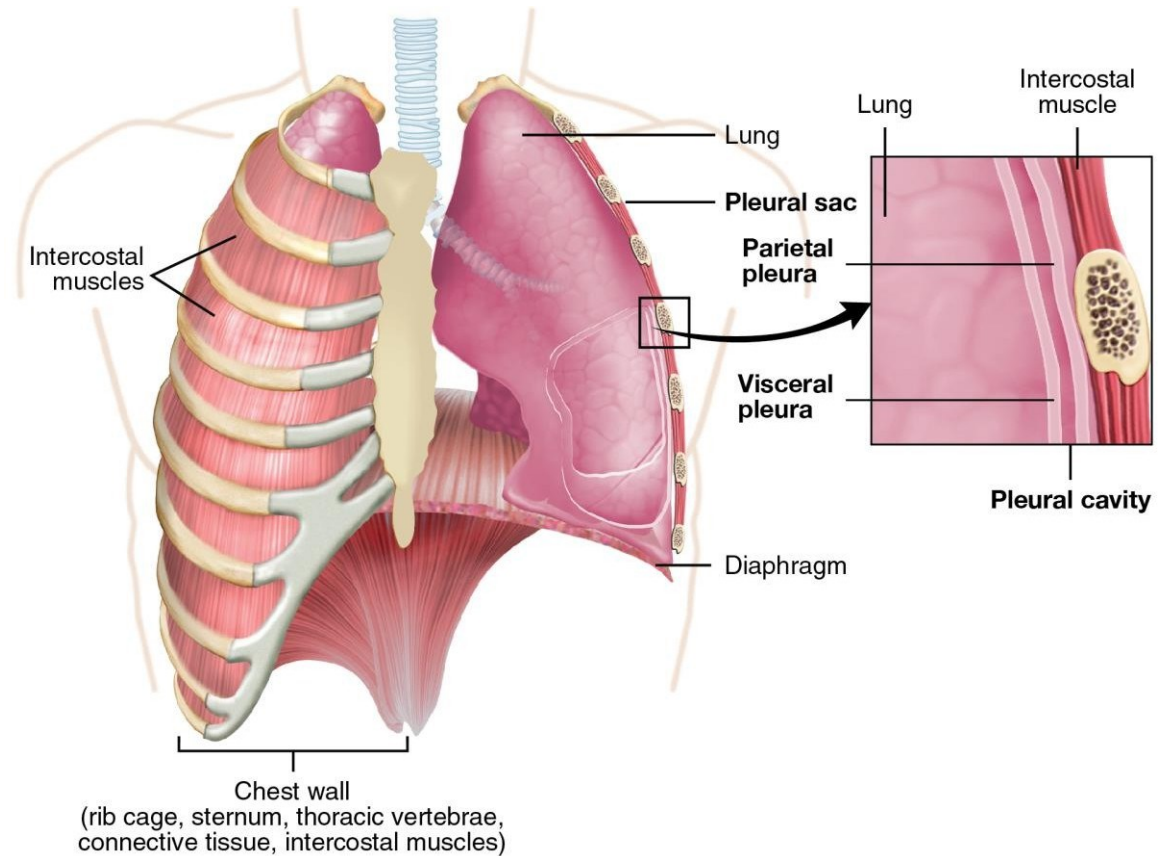


- Pleural effusion (definition, Types, pathogenesis, clinical presentation, diagnosis, management)
- Empyema (definition, Types, pathogenesis, clinical presentation, diagnosis, management)
- Hemothorax (definition, Types, pathogenesis, clinical presentation, diagnosis, management)
- Pneumothorax (definition, pathogenesis, clinical presentation, diagnosis, management)
- [Lung collapse] (definition, Types, pathogenesis, clinical presentation, diagnosis, management)

Introduction

Overview:

The pleura is a thin membrane that covers the entire surface of the lung (Visceral pleura), inner surface of the rib cage, diaphragm, and mediastinum (Parietal pleura).



Pleurisy

Pleurisy is an inflammatory condition that affects the pleura.

Etiology:

- **Viral** (Most common cause)

Adenovirus, coxsackieviruses, CMV, EBV, influenza virus, parainfluenza virus,

- **Bacterial**

Pneumonia (parapneumonic pleuritis), Tuberculosis

- **Inflammatory**

Uremia in renal failure, Systemic lupus erythematosus, Rheumatoid arthritis

- **Pulmonary** conditions

Pneumothorax , Asbestosis , Malignancy e.g., mesothelioma , Pulmonary embolism

- **Cardiac**

Myocardial infarction, Cardiac surgery

- **Drugs**

Amiodarone, bleomycin, methotrexate, Isoniazid, hydralazine

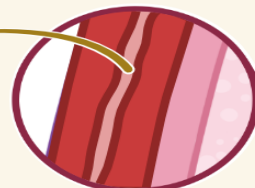
Pleurisy

BACKGROUND

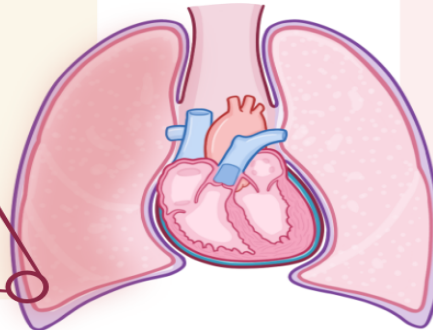
- * **VISCERAL PLEURA & PARIETAL PLEURA** BECOME INFLAMED
- * **PLEURAL CAVITY** USUALLY has a SMALL AMOUNT of FLUID
- * **INFLAMED MEMBRANES** RUB AGAINST EACH OTHER INSTEAD of GLIDING PAST
↳ STABBING CHEST PAIN & SHORTNESS of BREATH



NORMAL

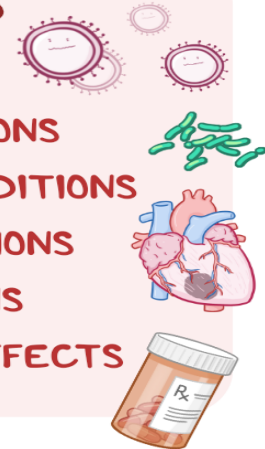


INFLAMED



CAUSES

- * **VIRAL INFECTIONS**
- * **BACTERIAL INFECTIONS**
- * **INFLAMMATORY CONDITIONS**
- * **PULMONARY CONDITIONS**
- * **CARDIAC CONDITIONS**
- * **MEDICATION SIDE EFFECTS**



SYMPTOMS

- * **PLEURITIC CHEST PAIN**
↳ WORSENS w/ COUGHING, SWALLOWING, or DEEP BREATHING
- * **HIGH-PITCHED BREATH SOUND** (PLEURITIC FRICTION RUB)
- * **DRY COUGH**
- * **SHORTNESS of BREATH**
- * **FEVER, CHILLS, & FATIGUE**



TREATMENT

- * **MANAGING ASSOCIATED PAIN & RESOLVING UNDERLYING CAUSE**
 - ~ NSAIDs for PAIN RELIEF
 - ~ ANTIBIOTICS for BACTERIAL INFECTIONS
 - ~ MEDS to PREVENT BLOOD CLOTS
 - ~ HOSPITALIZATION to DRAIN FLUID from LUNGS

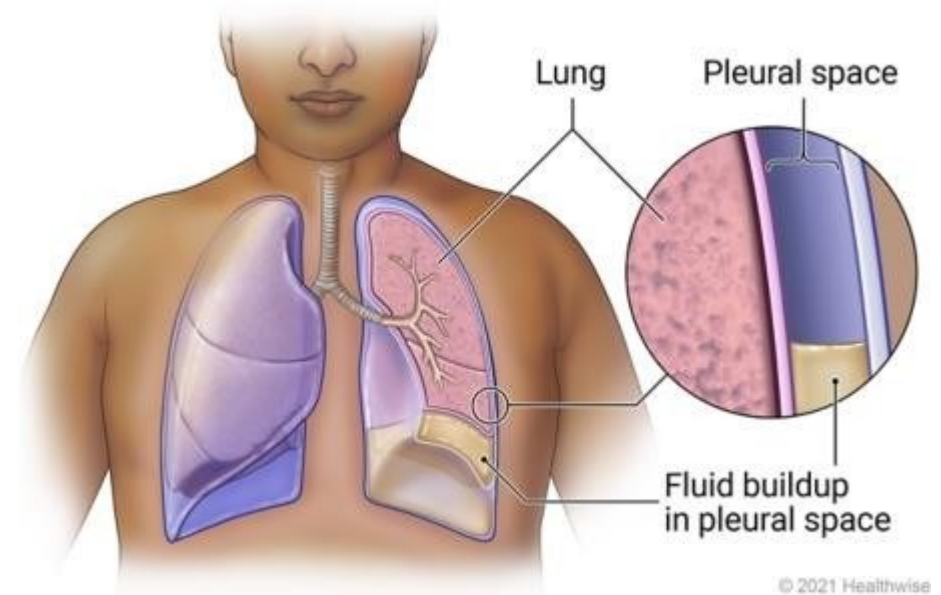
Pleural Effusion

Definition:

- An excessive amount of fluid between pleural layers that impairs the expansion of the lungs

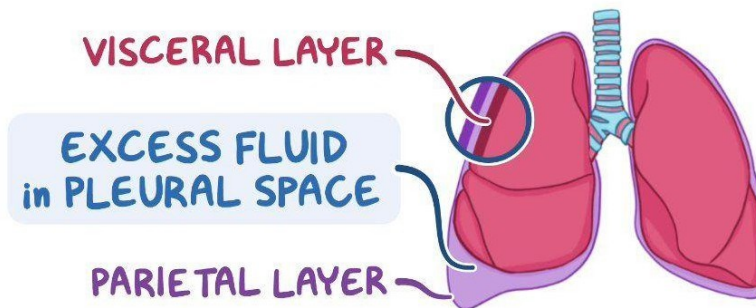
Causes:

- ↑ drainage of fluid into pleural space
- ↑ production of fluid by cells in the pleural space
- ↓ drainage of fluid from the pleural space
- Accumulation of blood is called hemothorax ,
Pus is called empyema, and
chyle is called chylothorax.



To Summarize

PLEURAL EFFUSION



CLINICAL MANIFESTATIONS

- * TROUBLE BREATHING
- * DRY COUGH
- * SHARP CHEST PAIN w/ INSPIRATION



TRANSUDATIVE EFFUSIONS

- * IMBALANCE of HYDROSTATIC & ONCOTIC PRESSURE in PULMONARY CAPILLARIES
- * FLUID → PLEURAL SPACE

EXUDATIVE EFFUSIONS

- * INFLAMMATORY DISORDERS CAUSE ↑ CAPILLARY PERMEABILITY
- * FLUID RICH in PROTEIN & LDH → PLEURAL SPACE



EMPYEMA / PYOTHORAX

- * FLUID is PURULENT

HEMOTHORAX

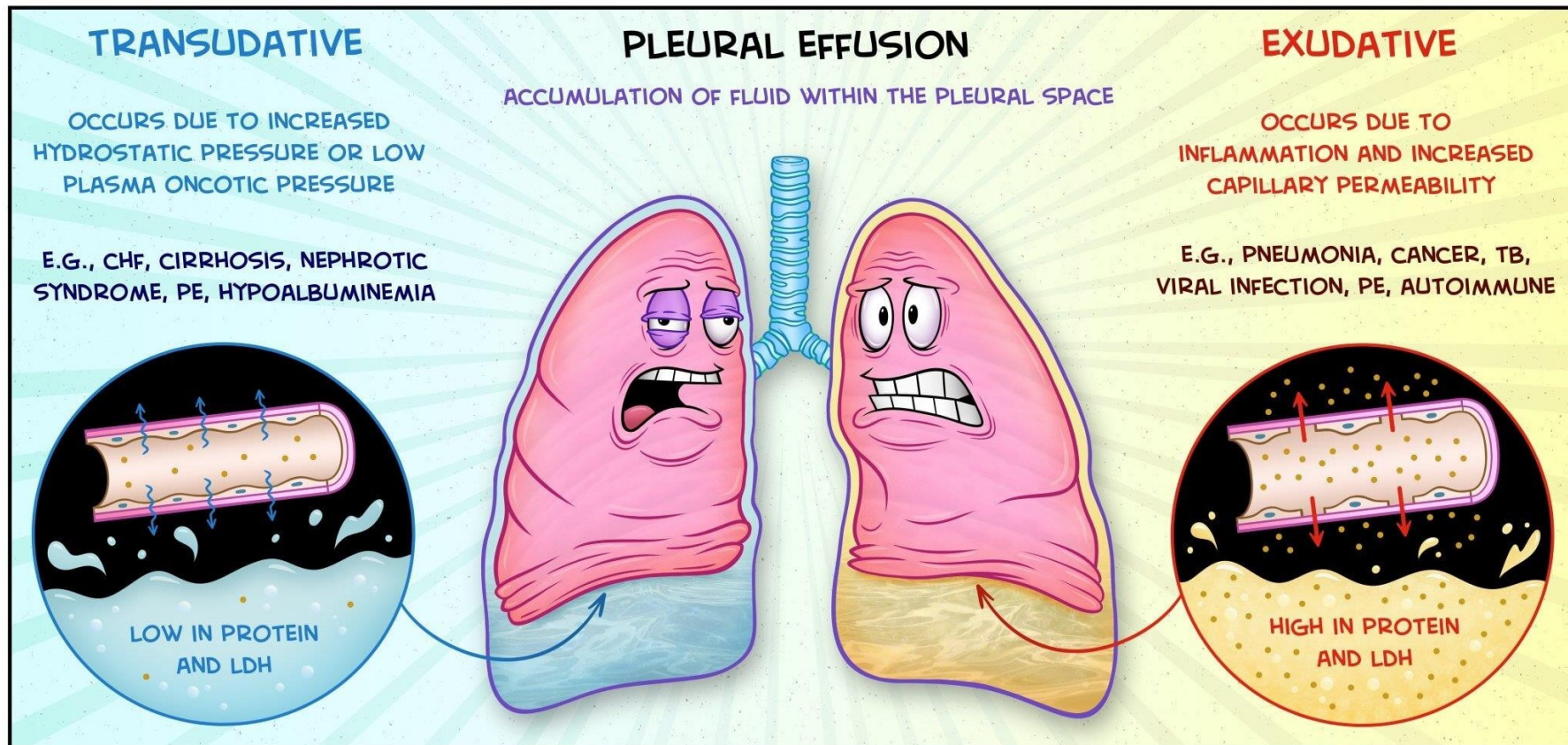
- * FLUID is SANGUINEOUS

CHYLOTHORAX

- * FLUID contains LYMPH

Pleural Effusion

Transudate Vs. Exudate:



Pleural Effusion

Transudate Vs. Exudate:

CLINICAL PEARL 2-4

Causes of Transudative and Exudative Pleural Effusions

Transudative

- CHF
- Cirrhosis
- PE
- Nephrotic syndrome
- Peritoneal dialysis
- Hypoalbuminemia
- Atelectasis

Exudative

- Bacterial pneumonia, TB
- Malignancy, metastatic disease
- Viral infection
- PE
- Collagen vascular diseases

Pleural Effusion

Empyema:

- Most common: pneumonia
- Less common
 - Infected hemothorax
 - Ruptured lung abscess
 - Esophageal tear
 - Thoracic trauma

Chyle:

- Trauma (including iatrogenic)
- Malignancy (e.g., lymphoma, bronchogenic carcinoma)
- Congenital lymphatic anomalies (e.g., lymphangiectasis)
- Thoracic duct obstruction

Hemothorax:

- Hemorrhagic blood disease
- Malignancy (Mesothelioma)

Pleural Effusion

Transudate Vs. Exudate:

Light's Criteria for Pleural Effusions		
	Transudate	Exudate
Protein (pleural/serum)	≤ 0.5	> 0.5
LDH (pleural/serum)	≤ 0.6 Pleural LDH $\leq$ two-thirds upper limit of normal serum LDH	> 0.6 Pleural LDH $>$ two-thirds upper limit of normal serum LDH

Pleural Effusion

Transudate Vs. Exudate:

TABLE 4.10. Pleural Fluid Analysis and Interpretation

PLEURAL FLUID TEST	INTERPRETATION
pH	If pleural pH is <7.2 with parapneumonic effusion, drainage is required
Hematocrit	>50% of peripheral hematocrit suggests hemothorax
Glucose	<60 mg/dL suggests a complicated parapneumonic effusion or malignancy; if especially low (<30), think RA
Triglycerides	>110 mg/dL points to chylothorax (thoracic duct lymph disruption); milky white; due to lymphoma, cancer, trauma, or lymphangioliomyomatosis
Lymphocytes	>50% lymphocytes is likely TB or malignancy
Eosinophils	>10% eosinophils is seen if air or blood is present, but most commonly due to pneumothorax ; also consider drug reaction, asbestos, paragonimiasis, Churg-Strauss syndrome, and pleural embolus
Amylase	↑ suggests acute pancreatitis, pancreaticopleural fistula or esophageal perforation

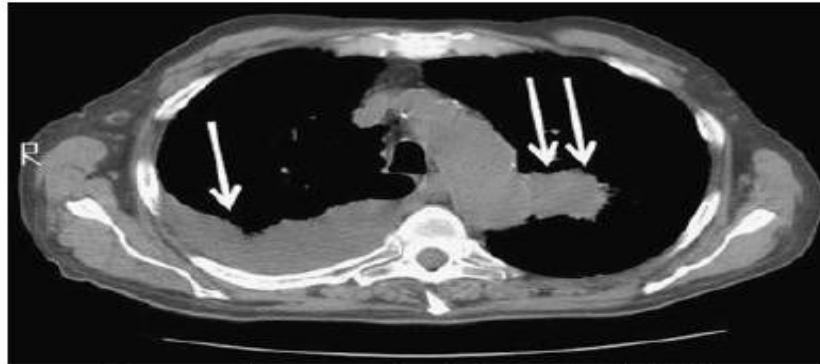
Pleural Effusion

Clinical Features:

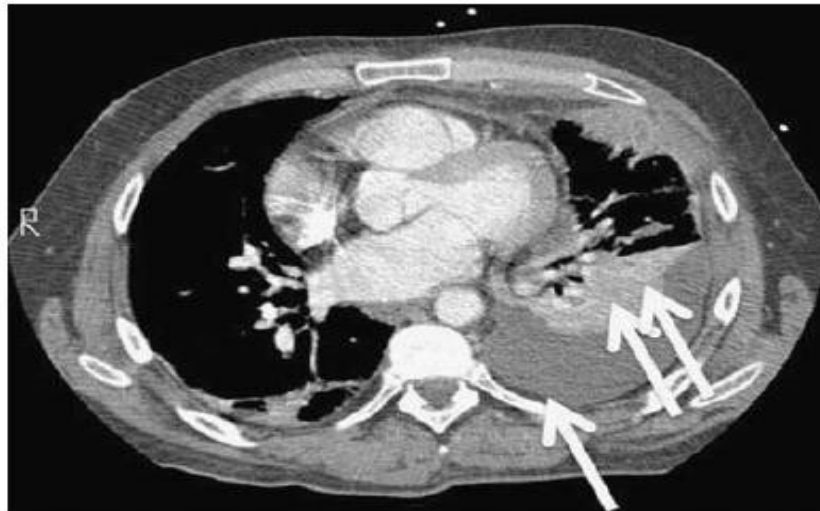
- Patients with a small pleural effusion (< 300 mL) are often asymptomatic
- Dyspnea
- Pleuritic chest pain
- Dry, nonproductive cough
- Symptoms of the underlying disease
- **Physical exam findings**
 - Inspection and palpation
 - Asymmetric expansion and unilateral lagging on the affected side
 - Reduced tactile fremitus due to fluid in the pleural space
 - Auscultation
 - Faint or absent breath sounds over the area of effusion
 - Percussion: dullness over the area of effusion

Pleural Effusion

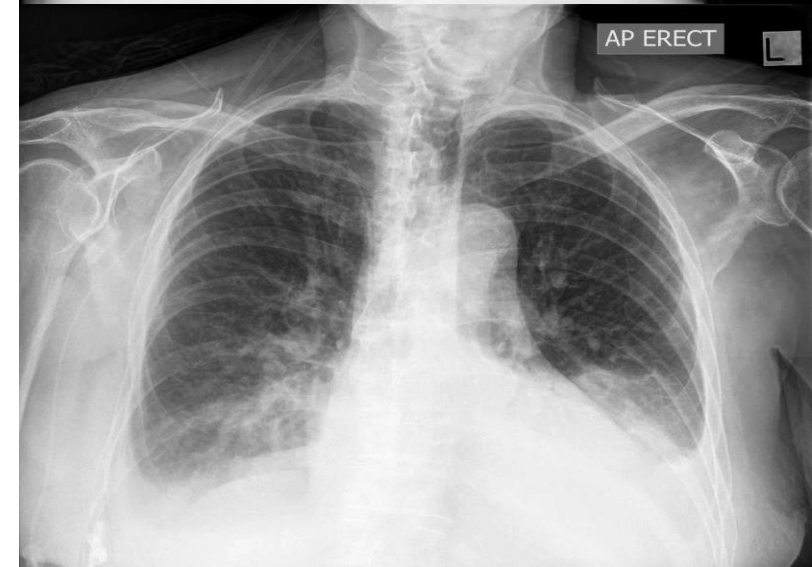
Diagnostics:
CXR (Look for blunting)



E-Fig. 19.2 Computed tomography of the patient in E-Fig. 19.1 shows a large, right pleural effusion (*arrow*) and left upper lobe mass (*double arrows*).



E-Fig. 19.3 Computed tomography demonstrates a pleural effusion (*arrow*) and atelectasis (*double arrows*).

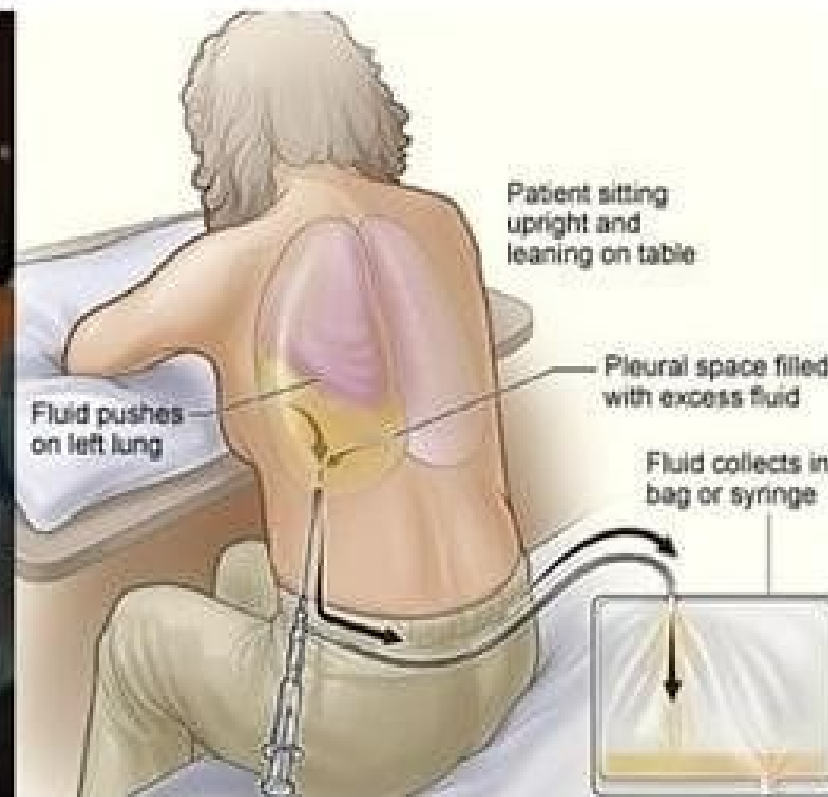


CT & US

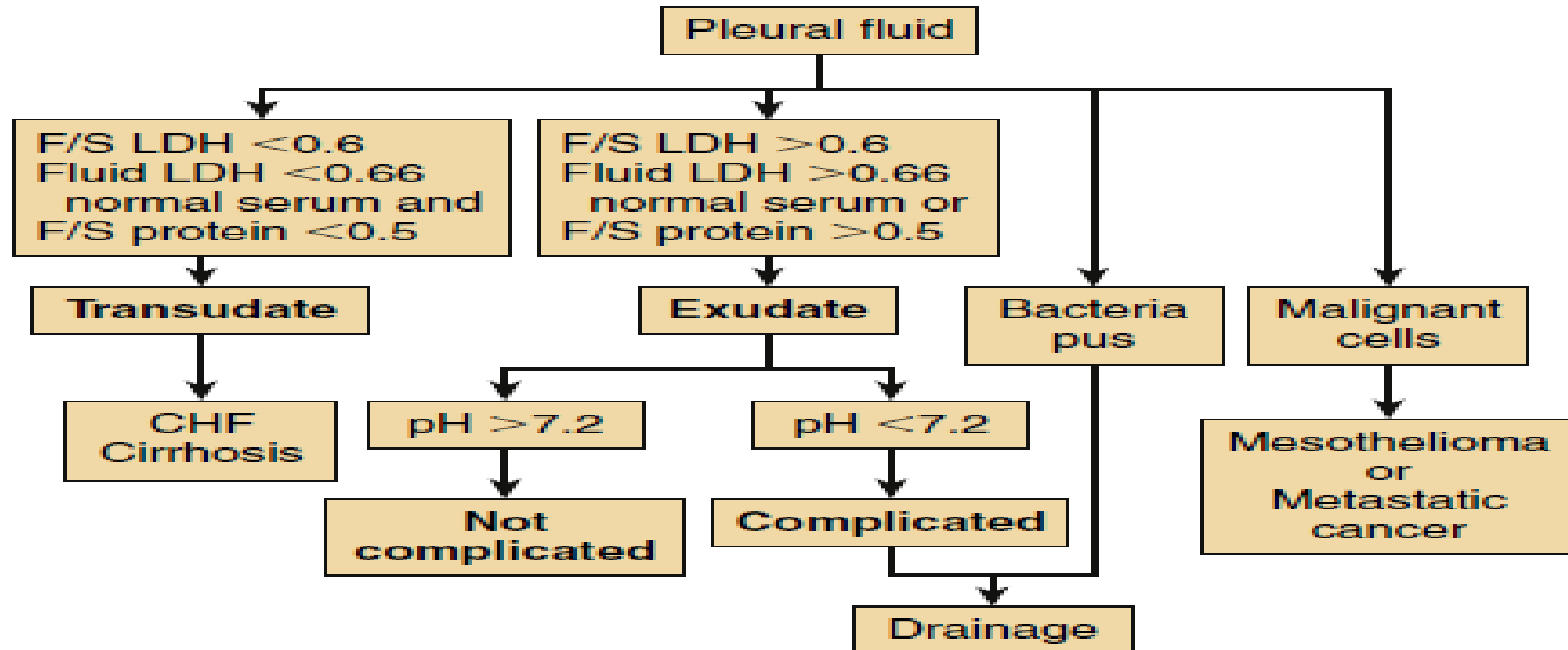
Pleural Effusion

Diagnostics:

- Thoracentesis is indicated in all new effusions
- Diagnostic Vs. Therapeutic



Dealing with pleural fluid analysis



E-Fig. 19.4 Work-up for pleural effusion. *CHF*, Congestive heart failure; *F/S*, fluid/serum; *LDH*, lactate dehydrogenase.

Pleural Effusion

Management:

1. Transudative effusions

- a. Diuretics and sodium restriction
- b. Therapeutic thoracentesis—only if massive effusion is causing dyspnea

2. Exudative effusions: treat underlying disease

3. Parapneumonic effusions

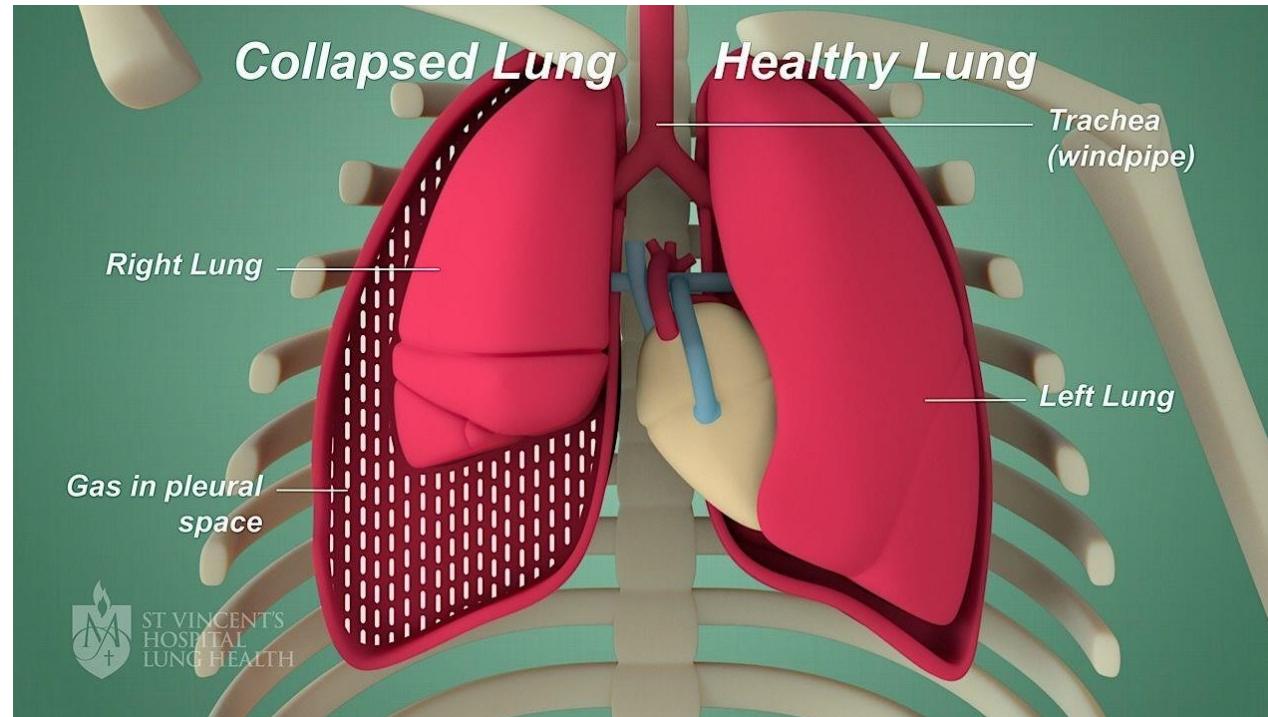
- a. Uncomplicated effusions: antibiotics alone
- b. Complicated effusions or empyema: Chest tube drainage & Intrapleural injection of fibrinolytic agents (sometimes)

Pneumothorax

Definition:

Pneumothorax is an accumulation of air in the pleural space.

If the accumulation is large enough, the underlying lung parenchyma may become collapsed and functionless.

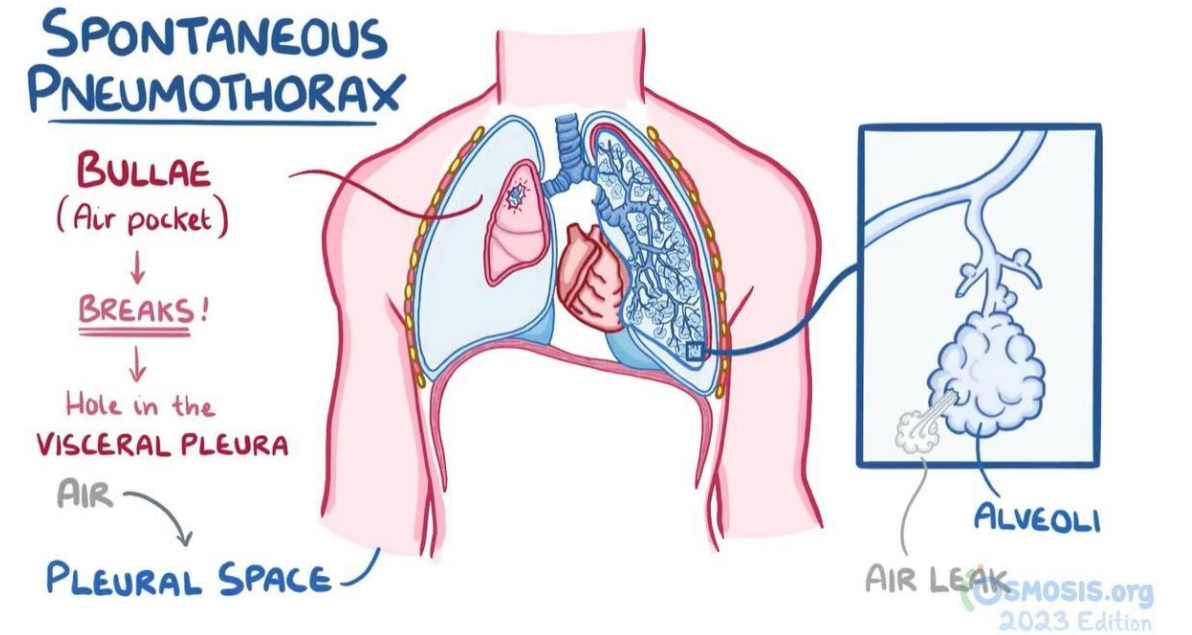


Pneumothorax

Classification:

Spontaneous pneumothorax

- **Primary** (no known lung disease)
 - seen in tall men who smoke cigarettes, Marfan syndrome, and cocaine use.
- **Secondary** (occurring with underlying lung disease, including COPD, CF, ILD, etc.)



Pneumothorax

Classification:

Traumatic pneumothorax:

- Penetrating or blunt trauma, or barotrauma from mechanical ventilation

Iatrogenic pneumothorax:

- thoracentesis, bronchoscopy, central venous catheter placement

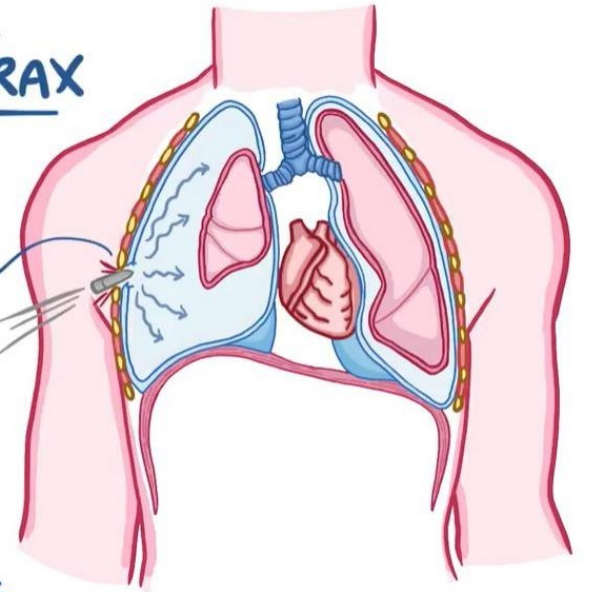
TRAUMATIC PNEUMOTHORAX

E.g. GUNSHOT
STAB WOUND

Rips through
PARIETAL
PLEURA

AIR

PLEURAL SPACE

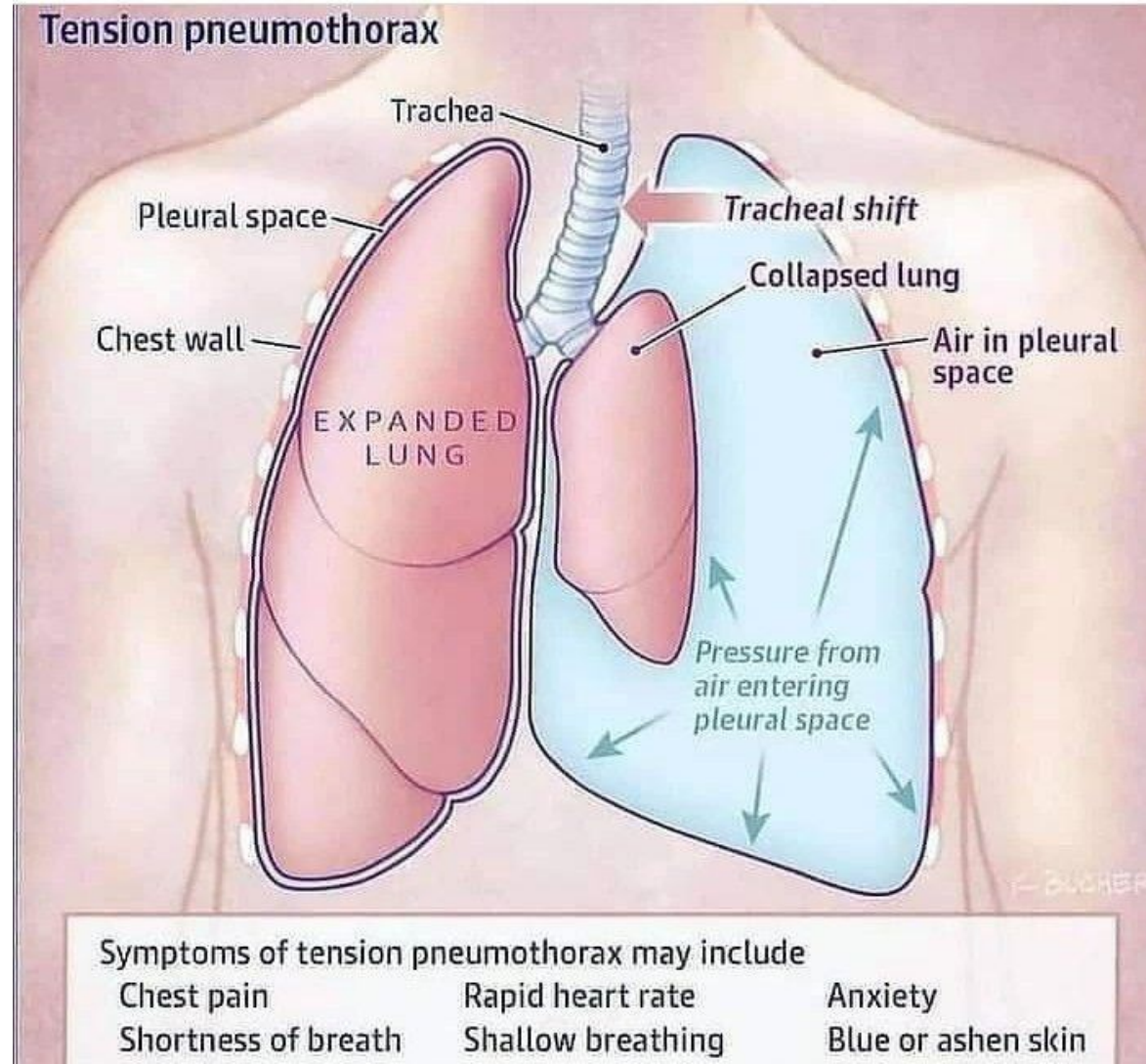


Pneumothorax

Classification:

Tension pneumothorax:

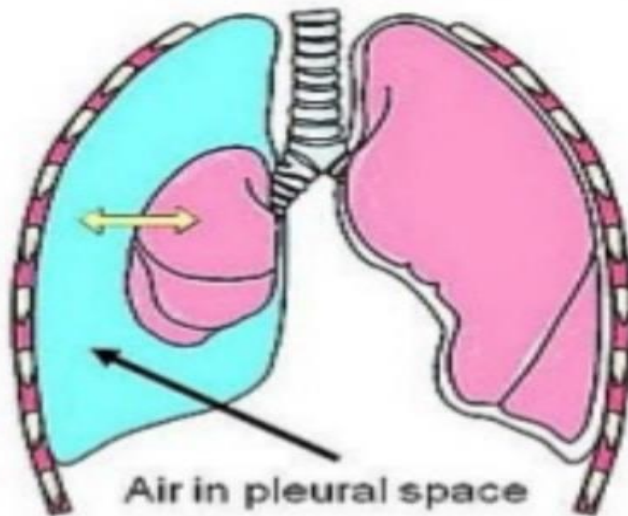
a life-threatening variant of pneumothorax characterized by progressively increasing pressure within the chest and cardiorespiratory compromise



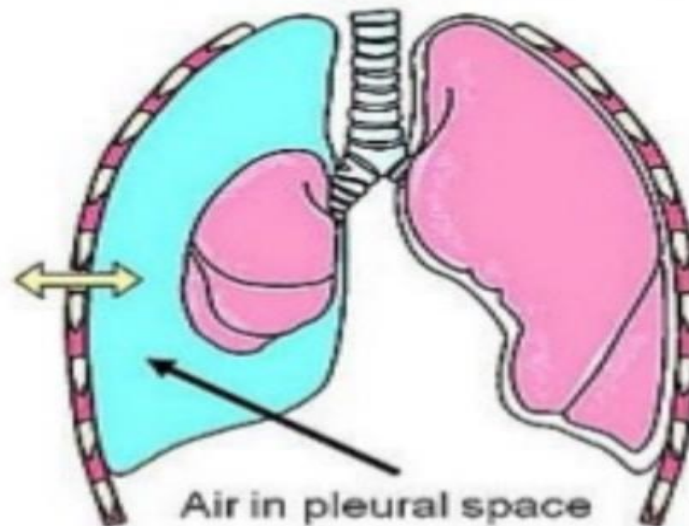
Pneumothorax

Pneumothorax

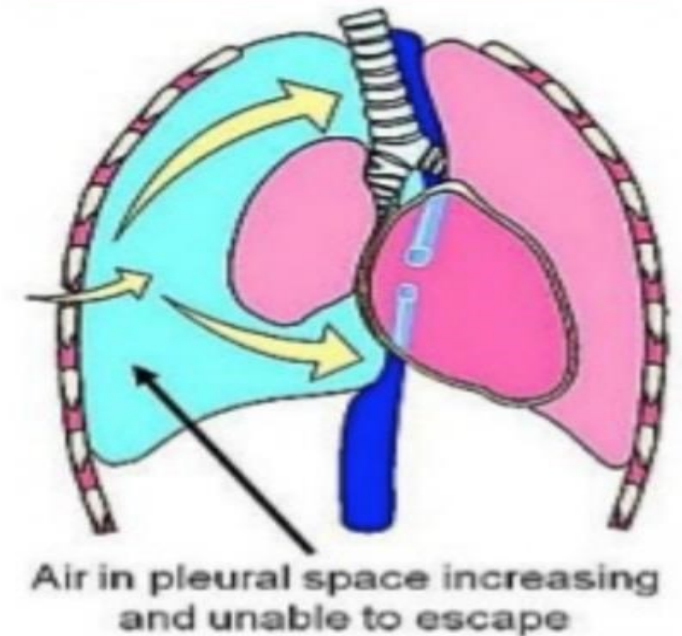
Closed pneumothorax



Open pneumothorax



Tension pneumothorax



Pneumothorax

- **Primary** (idiopathic or simple pneumothorax)

- Caused by ruptured **subpleural apical blebs**
- Risk factors
 - Family history, Male sex, Young age
 - Asthenic body habitus (**slim, tall stature**)
 - **Smoking** (90% of cases): up to 20-fold increase in risk (risk increases with the cumulative number of cigarettes smoked)

- **Secondary** (pneumothorax as a complication of underlying lung disease)

- **COPD** (smoking) → rupture of **bullae** in emphysema
- Infections
 - Pulmonary tuberculosis
 - Pneumocystis pneumonia → alveolitis, rupture of a cavity
- Cystic fibrosis → bronchiectasis with obstructive emphysema and bleb or cyst rupture
- Malignancy

Traumatic pneumothorax

- Blunt trauma (e.g., motor vehicle accident)
- Penetrating injury (e.g., gunshot, stab wound)
- Iatrogenic pneumothorax: mechanical ventilation

Pneumothorax

Clinical Features:

- Patients range from being asymptomatic to having features of hemodynamic compromise.
- Sudden, severe, and/or stabbing, ipsilateral pleuritic chest pain and dyspnea
- Reduced or absent breath sounds, hyperresonant percussion, decreased fremitus on same side
- Additional findings in tension pneumothorax
 - Severe acute respiratory distress: cyanosis, restlessness
 - Reduced chest expansion on the ipsilateral side
 - Distended neck veins and hemodynamic instability
 - Secondary injuries may be present (e.g., open or closed wounds).
 - Signs of tension pneumothorax in ventilated patients
 - Rapid decrease in SpO₂
 - Reduced air flow
 - Increased ventilation pressure
 - Skin emphysema

Pleuritic pain

Trachea deviation

Hyper resonance

Onset sudden

Reduced breath sounds (& dyspnea)

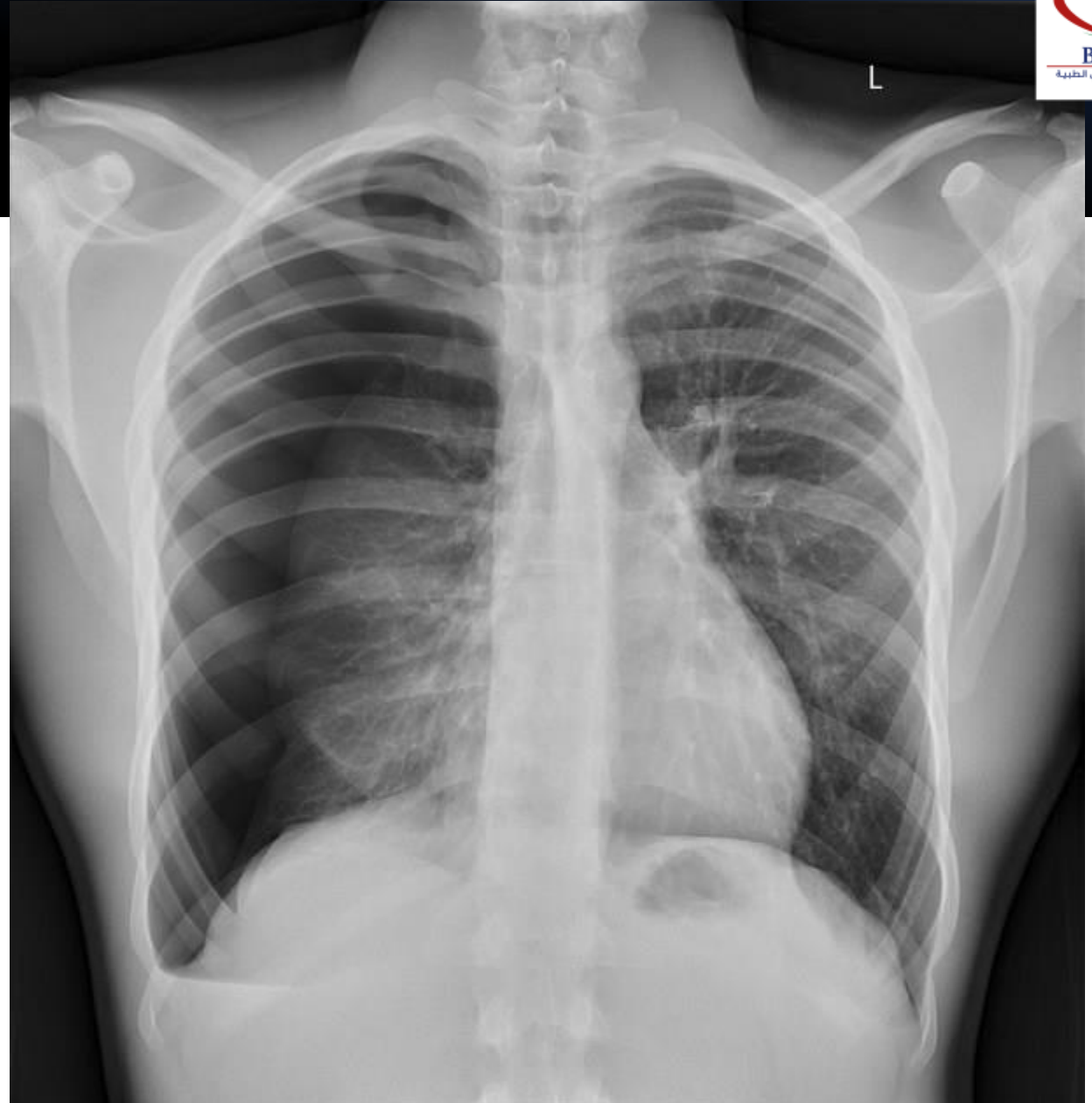
Absent fremitus (fremitus refers to vibratory tremors that can be felt through the chest by palpation)

X-ray shows collapse

Pneumothorax

Diagnostics:

- Clinical Diagnosis
- CXR to confirm
- POCUS
- CT Chest
- Tension pneumothorax is primarily a clinical diagnosis and prolonged diagnostic studies should be avoided in favor of initiating immediate treatment.



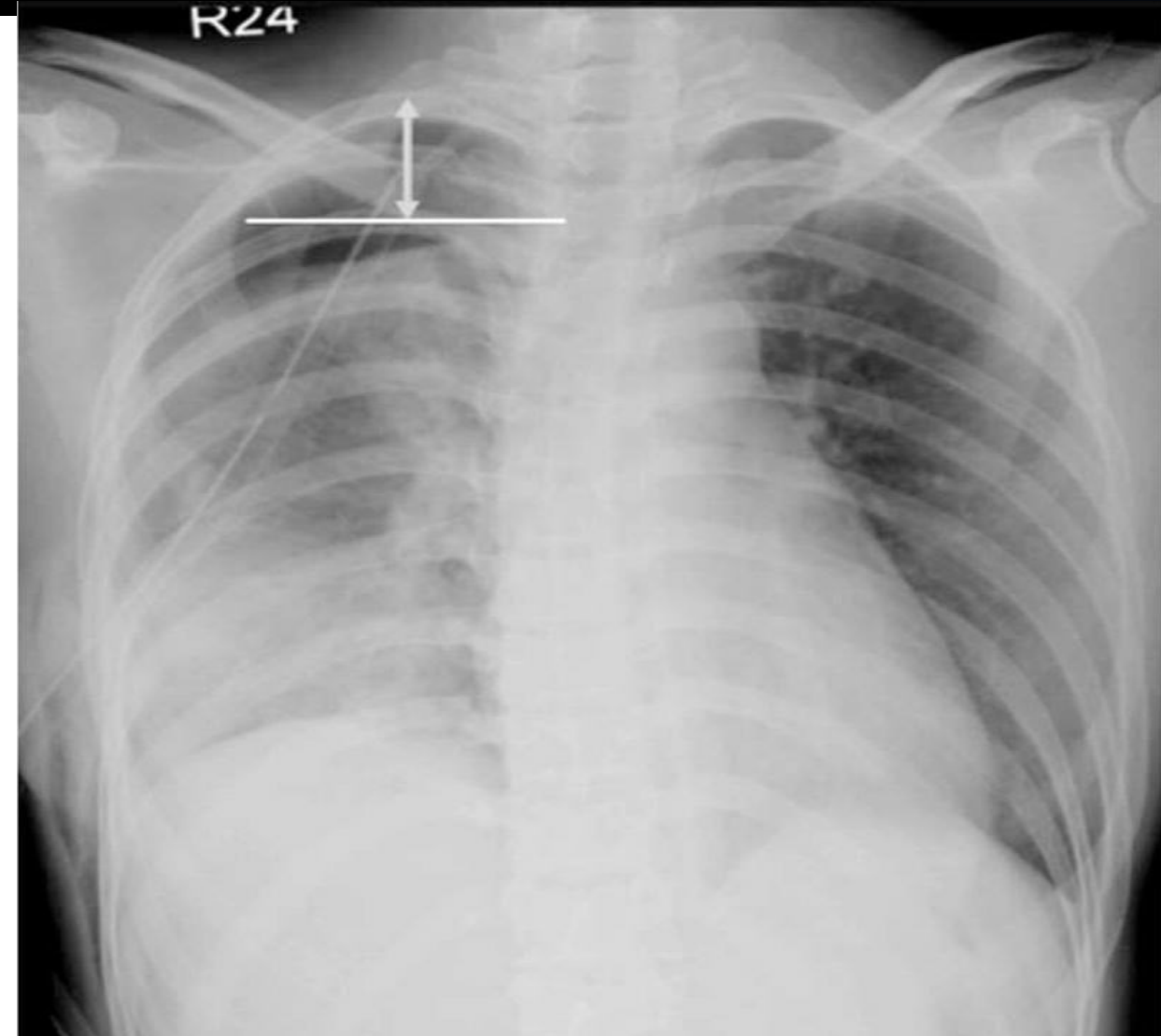
Pneumothorax

Diagnostics:

Determination of pneumothorax size

The size of a pneumothorax is assessed via imaging (e.g., CXR, CT chest).

How a pneumothorax is measured depends on regional guidelines, hospital policies, and personal preferences.



Pneumothorax

Management:

•All patients

- Assess patient stability
- Provide respiratory support and treat dyspnea.
- Evaluate the type and size of pneumothorax.

•**Unstable or high-risk patients:** e.g., with tension pneumothorax, bilateral pneumothorax, or who require mechanical ventilation

- Immediate chest decompression
- Treat obstructive shock if present.

Pneumothorax

Management:

Stable spontaneous pneumothorax management: depends on the risk of progression and recurrence

Low-risk: conservative management

Higher risk: chest tube placement

- **Small pneumothoraces:** Observation and O2 therapy.
- **Larger, more symptomatic pneumothoraces:** Drain with simple aspiration or a small-bore chest tube

Unstable (high risk) / Suspected tension pneumothorax

Emergency needle thoracostomy, followed immediately by chest tube placement

- **Traumatic pneumothorax management:** Most patients require chest tube placement

Pneumothorax

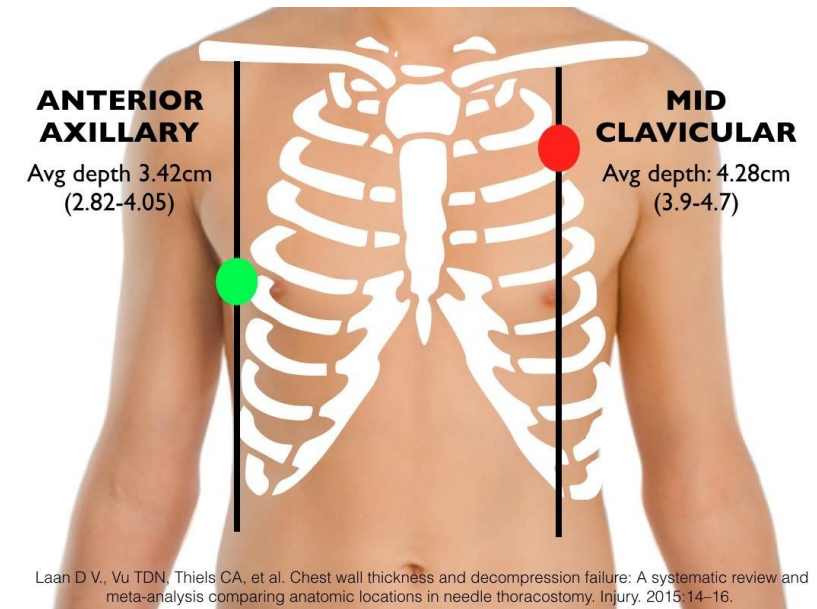
Needle thoracostomy Procedure:

- 2nd intercostal space at the midclavicular line
- In stable patients with a large (apex-to-cupola distance ≥ 3 cm)

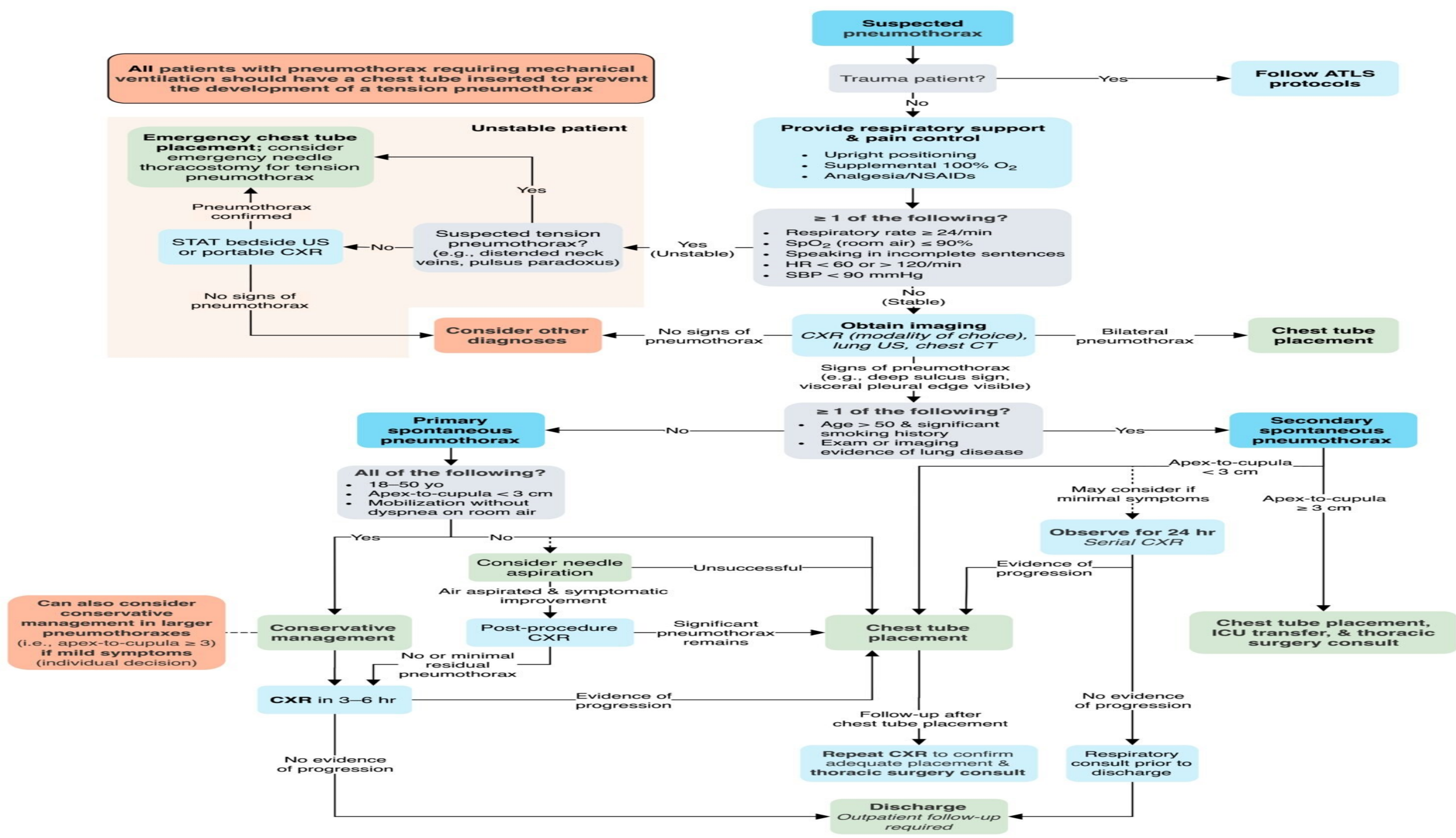
Chest tube placement Procedure

- 4th–5th intercostal space (nipple line), between the anterior and midaxillary line

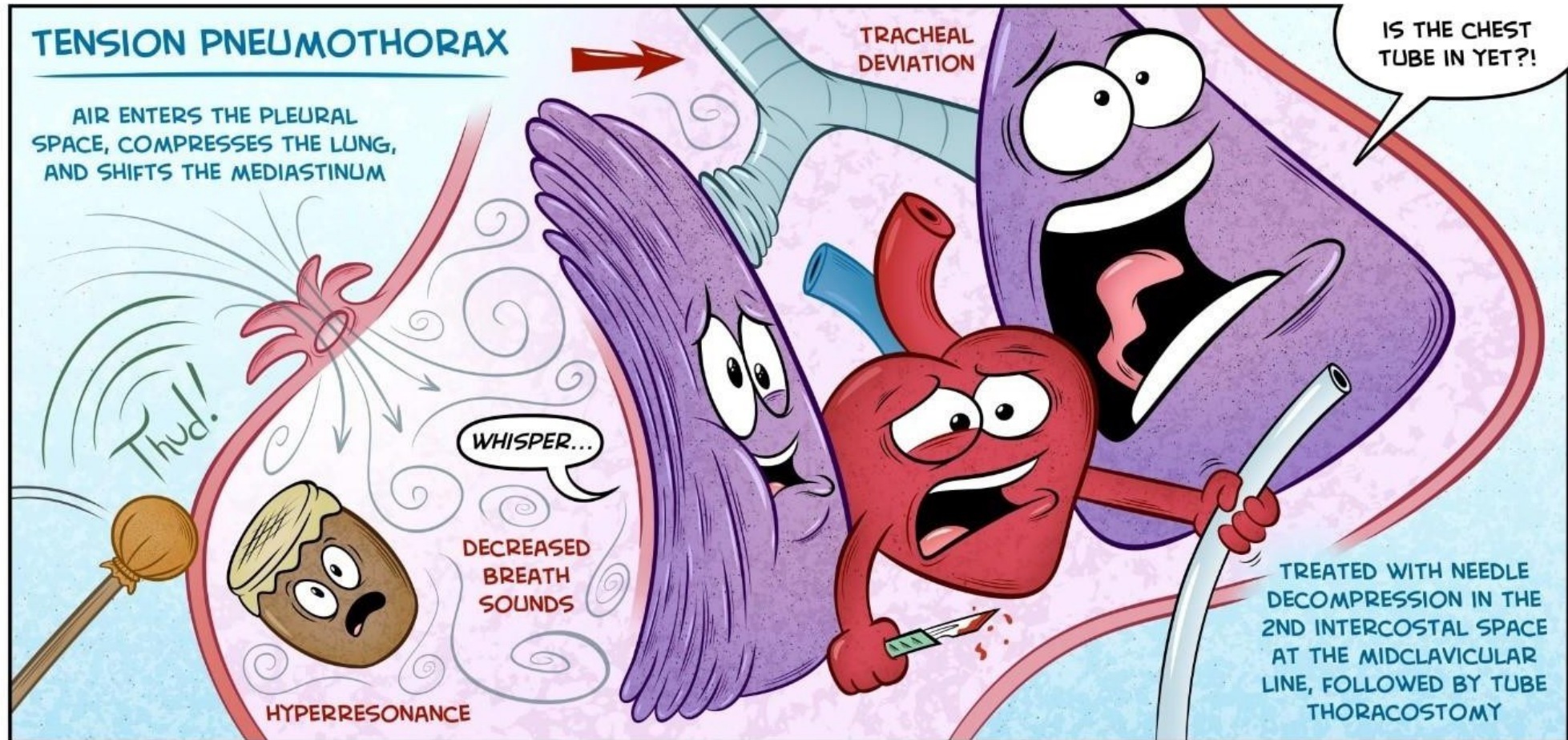
In Unstable patients / Apex-to-cupola distance ≥ 3 cm



All patients with pneumothorax requiring mechanical ventilation should have a chest tube inserted to prevent the development of a tension pneumothorax



To Summarize



Case Scenario 1

A 23-year-old man is brought to the emergency department 25 minutes after being involved in a high-speed motor vehicle collision in which he was the restrained driver. On questioning by the paramedics, he reported severe chest pain and mild dyspnea. On arrival, he is confused and unable to provide a history. His pulse is 93/min, respirations are 28/min, and blood pressure is 91/65 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 88%. He is able to move his extremities in response to commands. He opens his eyes spontaneously. Pupils are equal and reactive to light. Examination shows multiple bruises over the trunk and extremities. There is a 3-cm (1.2-in) wound at the left fifth intercostal space at the midclavicular line. There is jugular venous distention. Decreased breath sounds and hyperresonance on percussion are noted on the left. Which of the following is the most appropriate next step in management?

- a. CT scan of the chest
- b. Bronchoscopy
- c. Emergency thoracotomy
- d. Pericardiocentesis
- e. Needle decompression
- f. Transthoracic echocardiography
- g. X-ray of the chest

Case Scenario 2

A 49-year-old man comes to the hospital for a 10-day history of cough and worsening shortness of breath. He has sharp right-sided chest pain that worsens on inspiration and coughing. Two weeks ago, the patient was admitted to the hospital after passing out on the street from alcohol intoxication but he left against medical advice. He has coronary artery disease and hypertension. He takes no medications. He drinks 4 cans of beer daily and has smoked 2 packs of cigarettes daily for 20 years. His temperature is 38.5°C (101.3° F), pulse is 110/min, respirations are 29/min, and blood pressure is 110/65 mmHg. Examination shows poor dentition. There is dullness to percussion at the base of the right lung. Crackles and markedly decreased breath sounds are heard over the right middle and lower lung fields. An x-ray of the chest shows a right-sided loculated pleural effusion and consolidation of the surrounding lung with visible air bronchogram; there are no rib fractures. Thoracocentesis is performed. Examination of this patient's pleural fluid is most likely to show which of the following findings?

- a. Amylase of 200 U/L
- b. pH of 7.5
- c. Lymphocytosis of > 90%
- d. Pleural fluid LDH/serum LDH ratio of 0.5
- e. Hematocrit of 30%
- f. Lactate dehydrogenase of 45 U/L
- g. Glucose of 30 mg/dL
- h. Pleural fluid protein to serum protein ratio of 0.4

Questions?



References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine
20th edition
- Amboss
- Cecil Essentials of Medicine 10th edition
- First Aid for the internal medicine board
4th edition
- Osmosis

Thank You